

Exercise series 7:

FDs & BCNF



Problem 1

Consider the relation $R(A,B,C,D,E,F)$ with functional dependencies:

- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
- $\{A,B\} \rightarrow \{C\}$

- a) List all candidate keys of R .
- b) Is this relation in BCNF?
- c) Normalize the above relation schema to BCNF. Illustrate each step in the decomposition.



Problem 1

a) List all candidate keys of R.

- F must be part of all the keys because it can not be inferred.

R(A,B,C,D,E,F)

- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
- $\{A,B\} \rightarrow \{C\}$



Problem 1

a) List all candidate keys of R.

- F must be part of all the keys because it can not be inferred.
- Check {F}

R(A,B,C,D,E,F)

- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
- $\{A,B\} \rightarrow \{C\}$



Problem 1

a) List all candidate keys of R.

- F must be part of all the keys because it can not be inferred.
- Check {F}
- Check {A,F} {B,F} {C,F} {D,F} {E,F}

R(A,B,C,D,E,F)

- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
- $\{A,B\} \rightarrow \{C\}$



Problem 1

R(A,B,C,D,E,F)

- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
- $\{A,B\} \rightarrow \{C\}$

a) List all candidate keys of R.

- F must be part of all the keys because it can not be inferred.
- Check $\{F\}$
- Check $\{A,F\}$ $\{B,F\}$ $\{C,F\}$ $\{D,F\}$ $\{E,F\}$
- Check $\{A,E,F\}$



Problem 1

R(A,B,C,D,E,F)

- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
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a) List all candidate keys of R.

- F must be part of all the keys because it can not be inferred.
- Check $\{F\}$
- Check $\{A,F\}$ $\{B,F\}$ $\{C,F\}$ $\{D,F\}$ $\{E,F\}$
- Check $\{A,E,F\}$

b) Is this relation in BCNF?

- A superkey must contain a key, each key contains $\{F\}$. But $\{F\}$ is never on the left side. None of the FDs are in BCNF



Problem 1

R(A,B,C,D,E,F)

- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
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a) List all candidate keys of R.

- F must be part of all the keys because it can not be inferred.
- Check $\{F\}$
- Check $\{A,F\}$ $\{B,F\}$ $\{C,F\}$ $\{D,F\}$ $\{E,F\}$
- Check $\{A,E,F\}$

b) Is this relation in BCNF?

- A superkey must contain a key, each key contains $\{F\}$. But $\{F\}$ is never on the left side. None of the FDs are in BCNF

c) Normalize the above relation schema to BCNF.



- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D, E\}$
- $\{D\} \rightarrow \{A, B\}$
- $\{A, B\} \rightarrow \{C\}$

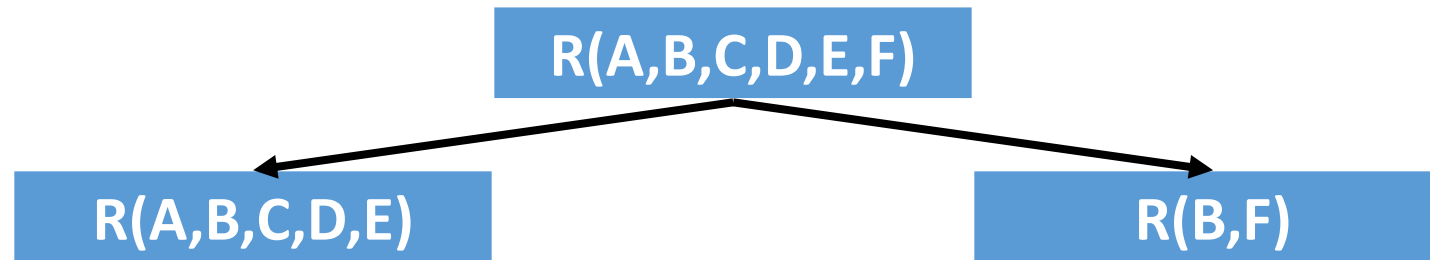
R(A,B,C,D,E,F)



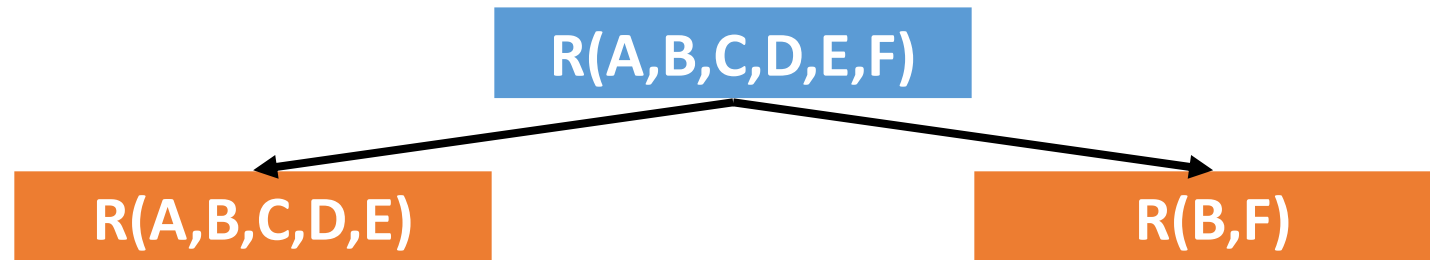
- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
- $\{A,B\} \rightarrow \{C\}$



- $\{B\} \rightarrow \{C\}$
- $\{B\}^+ = \{A,B,C,D,E\}$



- $\{B\} \rightarrow \{C\}$
- $\{C\} \rightarrow \{D,E\}$
- $\{D\} \rightarrow \{A,B\}$
- $\{A,B\} \rightarrow \{C\}$



DECOMPOSITION FOUND



Problem 2:

Consider the relation $R(A,B,C,D,E)$ with the functional dependencies:

- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$

- List all candidate keys of R .
- Is this relation in BCNF?
- Normalize the above relation schema to BCNF. Illustrate each step in the decomposition.



Problem 2:

a) List all candidate keys of R.

- $\{A,D\}$ must be part of all keys because it can not be inferred.

R(A,B,C,D,E)

- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$



Problem 2:

a) List all candidate keys of R.

- $\{A,D\}$ must be part of all keys because it can not be inferred.
- Check $\{A,D\}$

R(A,B,C,D,E)

- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$



Problem 2:

R(A,B,C,D,E)

a) List all candidate keys of R.

- {A,D} must be part of all keys because it can not be inferred.
- Check {A,D}

- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$

b) Is this relation in BCNF?

- {A,D} is the only key, and it is contained just in the first and last FD. So, no.



Problem 2:

R(A,B,C,D,E)

a) List all candidate keys of R.

- {A,D} must be part of all keys because it can not be inferred.
- Check {A,D}

- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$

b) Is this relation in BCNF?

- {A,D} is the only key, and it is contained just in the first and last FD. So, no.

c) Normalize the above relation schema to BCNF.



- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$

$R(A,B,C,D,E)$



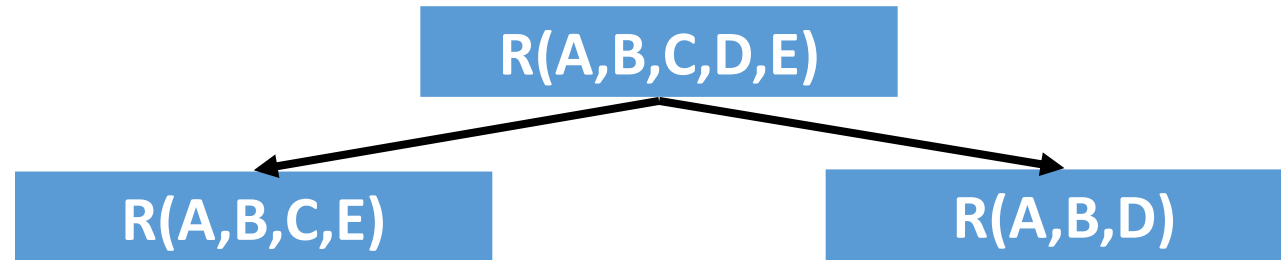
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- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$



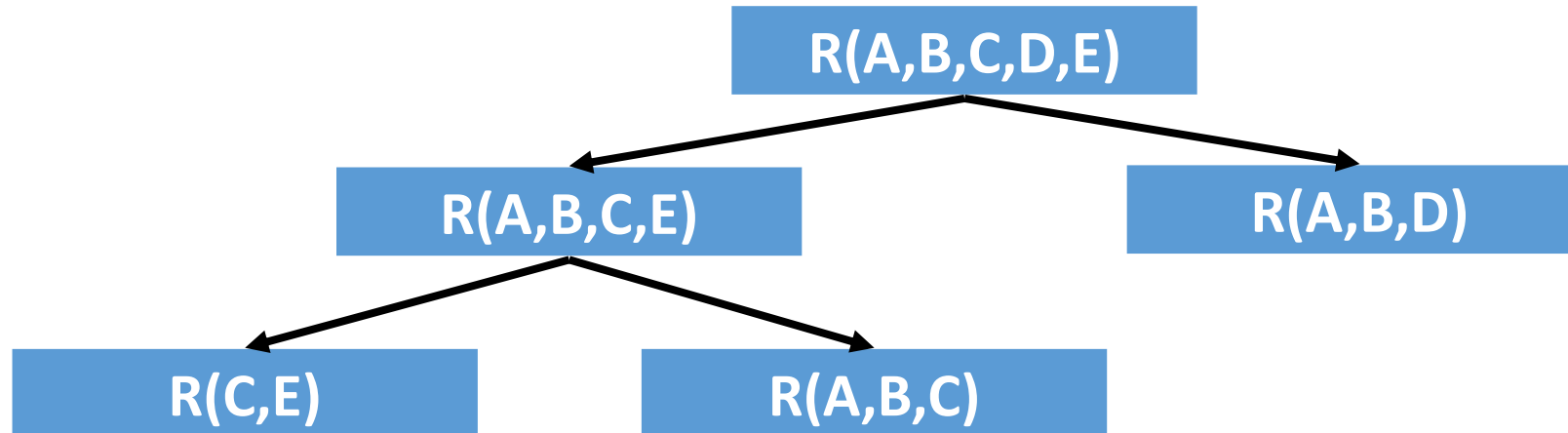
- $\{A,B\} \rightarrow \{C,E\}$
- $\{A,B\}^+ = \{A,B,C,E\}$



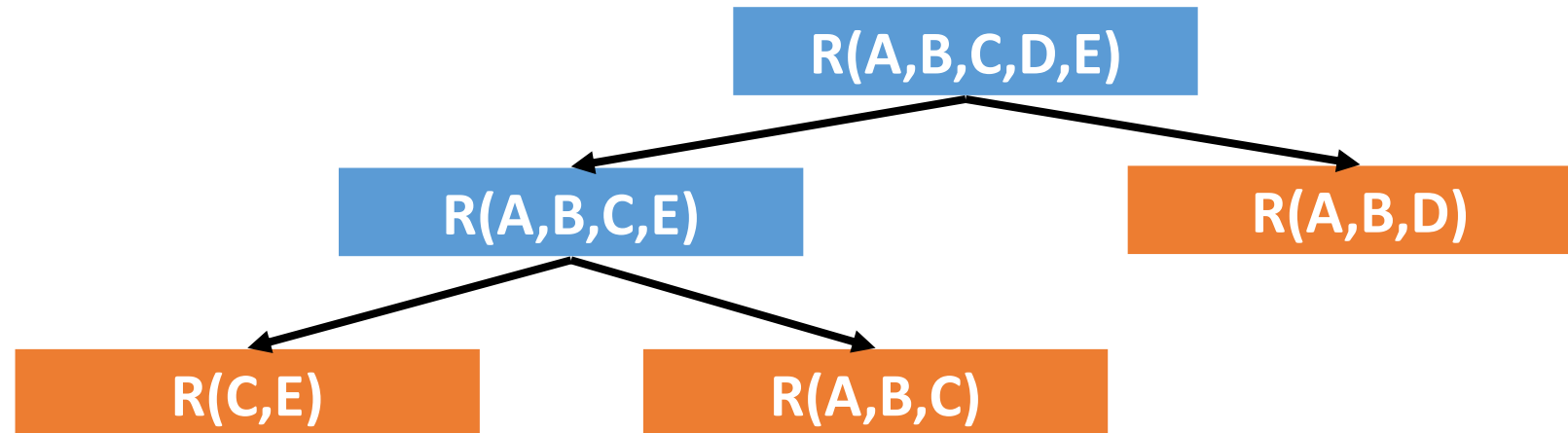
- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$



- $\{C\} \rightarrow \{E\}$
- $\{C\}^+ = \{C,E\}$



- $\{A,D\} \rightarrow \{B,C\}$
- $\{A,B\} \rightarrow \{C,E\}$
- $\{C\} \rightarrow \{E\}$
- $\{B,E\} \rightarrow \{C\}$
- $\{A,B,D\} \rightarrow \{E\}$



DECOMPOSITION FOUND



Problem 3:

Consider the relation $R(A,B,C,D,E)$ with the functional dependencies:

- $\{A\} \rightarrow \{B,C,E\}$
- $\{C,D\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{D\}$
- $\{E\} \rightarrow \{A\}$
- $\{B,C\} \rightarrow \{E\}$

Which of the following decompositions are *lossless*?

- a) $R_1=(A,B,C), R_2=(A,D,E)$
- b) $R_1=(A,B,C), R_2=(C,D,E)$
- c) $R_1=(A,B), R_2=(A,C), R_3=(B,C), R_4=(A,D,E)$



Problem 3:

R(A,B,C,D,E)

- $\{A\} \rightarrow \{B,C,E\}$
- $\{C,D\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{D\}$
- $\{E\} \rightarrow \{A\}$
- $\{B,C\} \rightarrow \{E\}$

a)	A	B	C	D	E
$R_1=(A,B,C)$	a	b	c	d_1	e_1
$R_2=(A,D,E)$	a	b_2	c_2	d	e

b)	A	B	C	D	E
$R_1=(A,B,C)$	a	b	c	d_1	e_1
$R_2=(C,D,E)$	a_2	b_2	c	d	e

c)	A	B	C	D	E
$R_1=(A,B)$	a	b	c_1	d_1	e_1
$R_2=(A,C)$	a	b_2	c	d_2	e_2
$R_3=(B,C)$	a_3	b	c	d_3	e_3
$R_4=(A,D,E)$	a	b_4	c_4	d	e

Problem 3:

R(A,B,C,D,E)	
•	$\{A\} \rightarrow \{B,C,E\}$
•	$\{C,D\} \rightarrow \{E\}$
•	$\{B\} \rightarrow \{D\}$
•	$\{E\} \rightarrow \{A\}$
•	$\{B,C\} \rightarrow \{E\}$

a)	A	B	C	D	E
$R_1=(A,B,C)$	a	b	c	d_1	e
$R_2=(A,D,E)$	a	b	c	d	e

b)	A	B	C	D	E
$R_1=(A,B,C)$	a	b	c	d_1	e_1
$R_2=(C,D,E)$	a_2	b_2	c	d	e

c)	A	B	C	D	E
$R_1=(A,B)$	a	b	c	d_1	e
$R_2=(A,C)$	a	b	c	d_2	e
$R_3=(B,C)$	a_3	b	c	d_3	e_3
$R_4=(A,D,E)$	a	b	c	d	e

Problem 3:

R(A,B,C,D,E)

- $\{A\} \rightarrow \{B,C,E\}$
- $\{C,D\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{D\}$
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a)	A	B	C	D	E
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$R_1=(A,B,C)$	a	b	c	d_1	e_1
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$R_4=(A,D,E)$	a	b	c	d	e

Problem 3:

R(A,B,C,D,E)

- $\{A\} \rightarrow \{B,C,E\}$
- $\{C,D\} \rightarrow \{E\}$
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c)	A	B	C	D	E
$R_1=(A,B)$	a	b	c	d_1	e
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$R_3=(B,C)$	a_3	b	c	d_3	e_3
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Problem 3:

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- $\{C,D\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{D\}$
- $\{E\} \rightarrow \{A\}$
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$R_1=(A,B,C)$	a	b	c	d_1	e_1
$R_2=(C,D,E)$	a_2	b_2	c	d	e

c)	A	B	C	D	E
$R_1=(A,B)$	a	b	c	d_1	e
$R_2=(A,C)$	a	b	c	d_2	e
$R_3=(B,C)$	a_3	b	c	d_3	e_3
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Problem 3:

R(A,B,C,D,E)

- $\{A\} \rightarrow \{B,C,E\}$
- $\{C,D\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{D\}$
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a)	A	B	C	D	E
$R_1=(A,B,C)$	a	b	c	d_1	e
$R_2=(A,D,E)$	a	b	c	d	e

b)	A	B	C	D	E
$R_1=(A,B,C)$	a	b	c	d_1	e_1
$R_2=(C,D,E)$	a_2	b_2	c	d	e

c)	A	B	C	D	E
$R_1=(A,B)$	a	b	c	d_1	e
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Problem 3:

R(A,B,C,D,E)

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a)	A	B	C	D	E
$R_1=(A,B,C)$	a	b	c	d_1	e
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$R_1=(A,B,C)$	a	b	c	d_1	e_1
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$R_1=(A,B)$	a	b	c	d_1	e
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$R_3=(B,C)$	a_3	b	c	d_3	e_3
$R_4=(A,D,E)$	a	b	c	d	e

Problem 4:

Consider the relation $R(A,B,C,D,E,F)$ with the functional dependencies:

- $\{A\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{F\}$
- $\{E,F\} \rightarrow \{C\}$

- List all candidate keys of R .
- Normalize the above relation schema to BCNF. Illustrate each step in the decomposition.



Problem 4:

R(A,B,C,D,E,F)

- $\{A\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{F\}$
- $\{E,F\} \rightarrow \{C\}$

a) List all candidate keys of R.

- $\{A,B,D\}$ must be part of all the keys because it can not be inferred.



Problem 4:

R(A,B,C,D,E,F)

- $\{A\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{F\}$
- $\{E,F\} \rightarrow \{C\}$

a) List all candidate keys of R.

- $\{A,B,D\}$ must be part of all the keys because it can not be inferred.
- Check $\{A,B,D\}$

b) Normalize the above relation schema to BCNF. Illustrate each step in the decomposition.

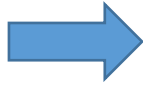


- $\{A\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{F\}$
- $\{E, F\} \rightarrow \{C\}$

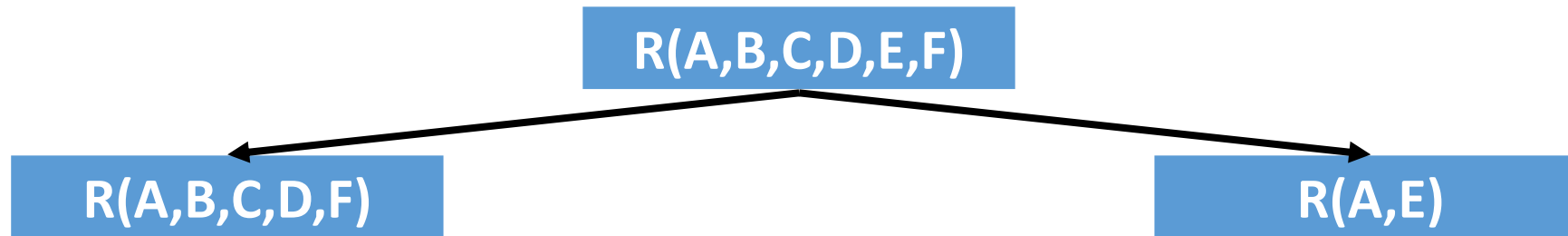
R(A,B,C,D,E,F)



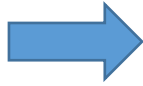
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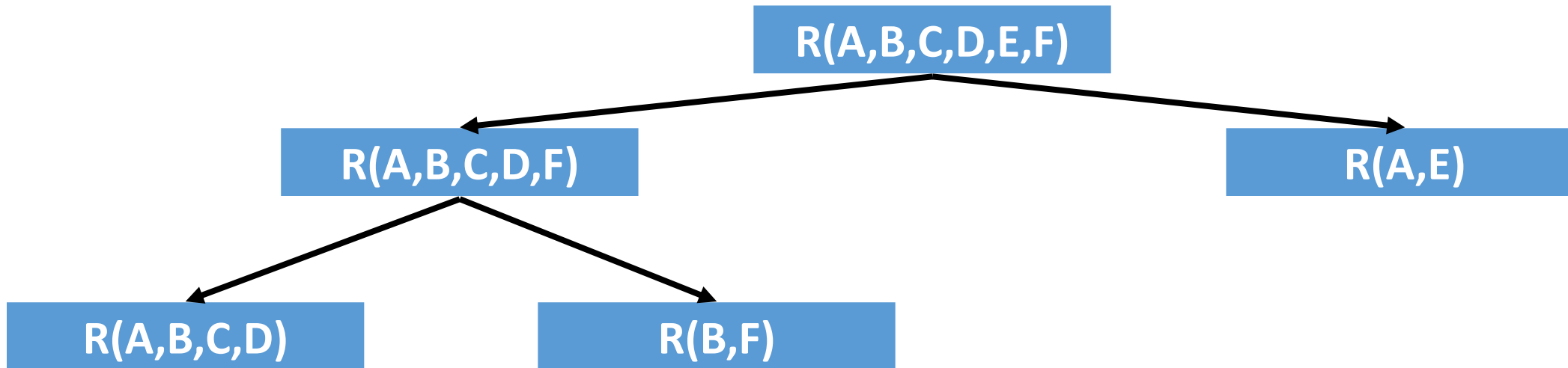
- $\{A\} \rightarrow \{E\}$
- $\{A\}^+ = \{A, E\}$



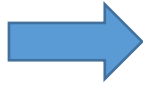
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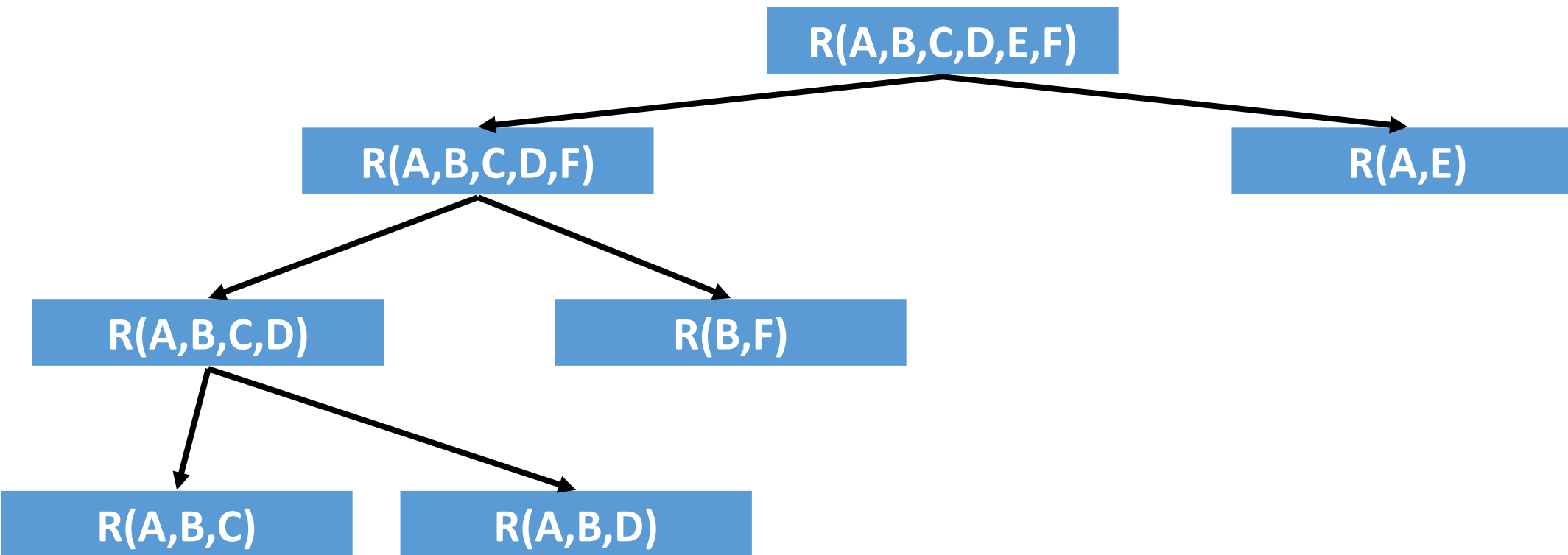
- $\{B\} \rightarrow \{F\}$
- $\{B\}^+ = \{B, F\}$



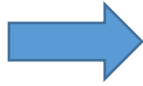
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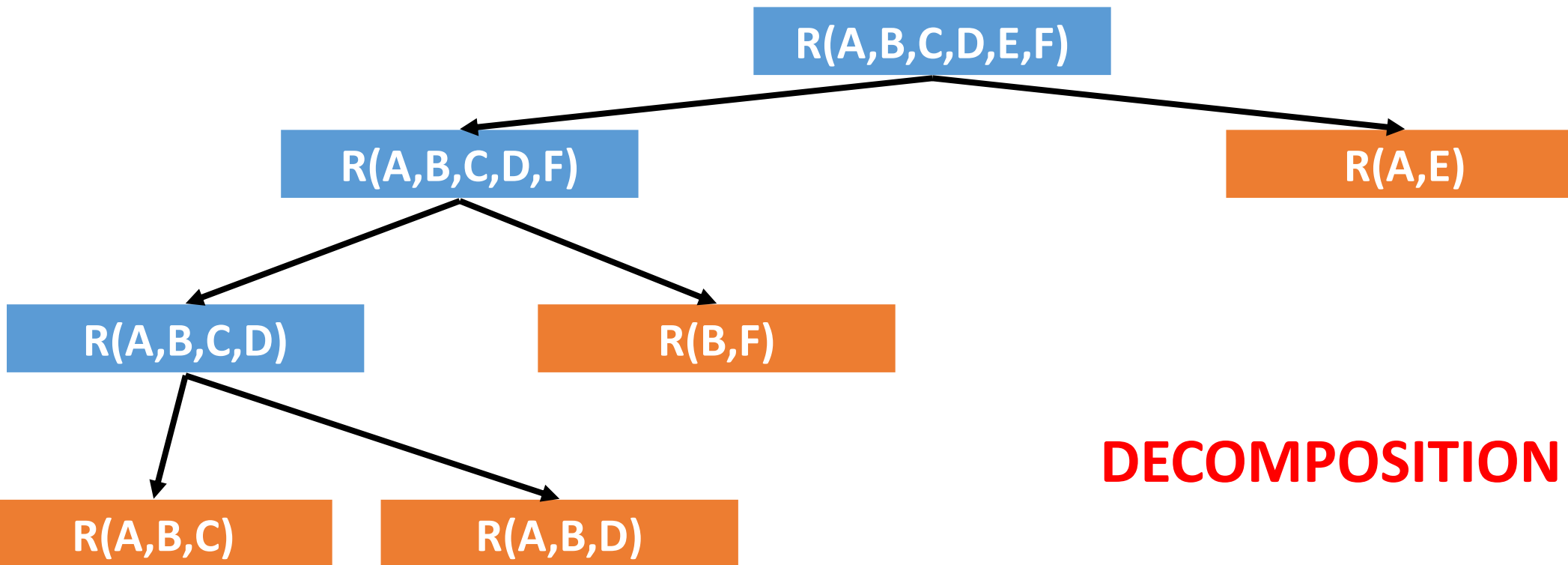
- $\{A, B\} \rightarrow \{C\}$
- $\{A, B\}^+ = \{A, B, C, E, F\}$



- $\{A\} \rightarrow \{E\}$
- $\{B\} \rightarrow \{F\}$
- $\{E, F\} \rightarrow \{C\}$



- $\{A, B\} \rightarrow \{C\}$
- $\{A, B\}^+ = \{A, B, C, E, F\}$



Problem 5: Exam question 2019 (almost)

Consider the following database relation that stores data about employees. Every employee has a unique identifier. Furthermore, for every employee his or her name, gender, address, bank account and salary are stored. Every employee works for exactly one department and is working on one or more projects. Each project has a unique project name and a project manager, who is also an employee. This data is stored into one relation. The following example shows how it is stored:

id	name	gender	address	dept	account	salary	proj	manager-ID
1	Bert	M	Sesame St. 1	IT	010234	2200	Alpha	1
1	Bert	M	Sesame St. 1	IT	010234	2200	Beta	4
2	Ernie	M	Sesame St. 1	Accounting	010345	2400	Delta	3
3	Ruth	F	NE Blvd. 24	Accounting	027849	3200	Delta	3
4	Mary	F	SW Blvd. 35	IT	027682	3500	Beta	4
4	Mary	F	SW Blvd. 35	IT	027682	3500	Delta	3
5	Donald	M	White House	Marketing	000000	2800	MAGA	5

As you can see, employee Bert is working on two projects. Hence, two tuples are added in the relation to store Bert's involvement into both projects. The other attributes which are unique for this employee are hence duplicated (name, gender, address, department, account, salary). The same holds for Mary.

Problem 5:

In this relation, the following rules hold:

1. Every employee has a unique ID. Employees have one name, gender, address, department, account, salary.
2. Every project has exactly one manager.
3. Addresses are not necessarily unique; there may be multiple employees that share the same address. It is, however, forbidden for two employees with the same address to work at the same department (e.g. to avoid nepotism).
4. When a person works on multiple projects, multiple tuples are stored for this person, one for each project and its project manager. The other attributes are duplicated for the employee.
5. The account of a person is unique; every person has one account, and no two employees may have the same account (e.g., due to legal restrictions salaries need to be paid on a personal account).

Question:

- a. For each of the rules 1-5, if possible, state it as a functional dependency. If this is not possible, explain *why* this is not possible.
- b. List all candidate keys of the relation.
- c. Is this relation in BCNF? If not decompose it.



Problem 5:

- a) For each of the rules 1-5, if possible, state it as a functional dependency. If this is not possible, explain *why* this is not possible.
1. $\text{id} \rightarrow \text{name, gender, address, dept, account, salary}$
 2. $\text{proj} \rightarrow \text{manager-ID}$
 3. $\text{address, dept} \rightarrow \text{id}$
 4. Not possible
 5. $\text{account} \rightarrow \text{id}$
- b) List all candidate keys of the relation: $\{\text{proj, id}\}$ $\{\text{proj, account}\}$ $\{\text{proj, address, dept}\}$
- c) Is this relation in BCNF? If not decompose it.

id	name	gender	address	dept	account	salary
----	------	--------	---------	------	---------	--------

id	proj
----	------

manager-ID	proj
------------	------

